

Lokesh Singh

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PROJECTS

1. LokiAI — Local AI System with Custom Vector Database & RAG | *Python · FastAPI · Ollama (LLaMA 3.2) · Custom VectorDB · HNSW · KDTree · RAG Pipeline · Semantic Search* [\[Github Link\]](#)

- ▶ **Developed a fully offline AI inference system** using Python FastAPI and Ollama (LLaMA 3.2), building a **custom in-memory vector database from scratch** implementing 3 search algorithms (HNSW, KDTree, Brute Force) and 3 distance metrics (Cosine Similarity, Euclidean, Manhattan) — no cloud dependency or paid APIs required.
- ▶ **Implemented a complete end-to-end RAG (Retrieval-Augmented Generation) pipeline** covering document ingestion, embedding generation via Ollama, top-K semantic search for context retrieval, and LLM-powered document Q&A, enabling natural language querying over user-uploaded data.
- ▶ **Integrated PCA vector space visualization** for real-time embedding exploration, and wrote a **smoke test suite** validating embedding, ingestion, and semantic search correctness across all algorithm-metric combinations with structured API endpoints (FastAPI REST API).
- ▶ **Analyzed high-dimensional embedding datasets** and implemented **efficient semantic search** techniques for **large-scale data retrieval and analytics**.

2.PortBot - AI Portfolio Chatbot *Python · FastAPI · Groq API· Prompt Chaining· Pydantic(Structured output) ·Streaming* [\[Github Link\]](#)

- ▶ **Developed a full-stack AI portfolio chatbot** using Python FastAPI, React, and the Groq LLM API that lets recruiters interview a **personal AI representative** in natural language — answering strictly from a structured JSON profile of education, skills, projects, experience, and achievements.
- ▶ **Implemented an end-to-end conversational pipeline** covering profile loading, grounded prompt construction, Groq LLM inference, and **token-by-token response streaming** via a FastAPI REST API and the React Fetch API, with multi-turn memory so follow-ups like "which project was the hardest?" resolve correctly.
- ▶ **Integrated job-description matching** that returns candidate fit, skill gaps, and strengths as **structured JSON**; built a **responsive React chat UI** with auto-scroll, dark mode.

ACHIEVEMENTS

- ▶ 🏆 **Orange Promptathon 2026 — Winner:** Built and deployed a winning HVAC Quality Inspection System .
- ▶ 🏆 **SunHacks Nashik Hackathon 2025 — Top 10** (out of **600+** competing teams globally).

PROGRAMMING SKILLS

- **Languages** : Java,SQL, Python
- **Cloud Fundamentals:** Linux(Commands & Basic Shell Scripting),AWS(EC2,VPC,IAM,S3,RDS),Docker
- **Tools:**Git,Github,PostMan
- **Backend:**SpringBoot,NodeJs, FastAPI,REST APIs,JWT Authentication
- **Databases** : QDRANT, MySQL, MongoDB.
- **AI Skills:** Scikit-learn, Pandas, NumPy, ReAct,Prompt Chaining, RAG Evaluation,Langgraph,HNSW.
- **Core Fundamentals** : Database Management System, Object-Oriented Programming, Operating System, Data Structures and Algorithms,Computer Networks.

EDUCATION

Shri Ramdoebaba College of Engineering and Management CGPA:8.39	Graduation Year:2027
St.Xavier’s High School,Nagpur HSC(XII)	Percentage:78.4%
St.Xavier’s High School,Nagpur SSC(XII)	Percentage:92%

